

Topics in CS: Problem Set 4

Due date: November 30, 2025.

Question 1. (20 points)

1. Write the following permutations in S_5 as $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ \sigma(1) & \sigma(2) & \sigma(3) & \sigma(4) & \sigma(5) \end{pmatrix}$:

(a) $(1254)(134)(245)$.

(b) $(1254)^{-1}(1342)(1254)$.

2. Consider $\sigma = (123)(3456)(5678)(891)(3456)(67)(789) \in S_9$.

(a) Write τ as a product of disjoint cycles.

(b) What is the parity of σ .

Question 2. (10 points) Prove whether the following subsets of S_4 are subgroups of S_4 .

1. $H_1 = \{\sigma \in S_4 \mid \sigma(1) = 3\}$.

2. $H_2 = \{\sigma \in S_4 \mid \sigma(2) = 2\}$.

Question 3. (20 points)

1. Find a subgroup of S_4 that is isomorphic to the multiplicative group $\{1, -1, i, -i\} \subseteq \mathbb{C}^*$.

2. Prove or refute the following claim:

Let $n \in \mathbb{N}$ such that $n \geq 2$ and let G be a commutative subgroup of S_n . Then, G is cyclic.

Hint: check what are the commutative subgroups for small values of n (e.g., $n = 2, 3, 4$), and see if you can identify a pattern.

Question 4. (20 points) Let G be a group and let H, K be subgroups of G .

Define $HK = \{hk \mid h \in H, k \in K\}$.

1. Consider the function $f : H \times K \rightarrow HK$ defined by $f(h, k) = hk$. Prove that for every $a \in HK$ the size of the preimage $f^{-1}(a) = \{(h, k) \in H \times K \mid hk = a\}$ is $|H \cap K|$.

Hint: find a bijection between $H \cap K$ and $f^{-1}(a)$.

2. Prove that for every $a, b \in HK$, if $a \neq b$ then $f^{-1}(a) \cap f^{-1}(b) = \emptyset$.

3. Conclude that $|HK| = \frac{|H| \cdot |K|}{|H \cap K|}$.

4. Prove or refute the following claim: HK is a subgroup of G .

Question 5. (20 points) Denote by G the group $GL_2(\mathbb{Z}_2) = \{A \in M_2(\mathbb{Z}_2) \mid \det(A) \neq 0\}$.

1. What is $|G|$.
2. Write down all the cyclic subgroups of G .
3. Let $n = |G|$ and consider the natural bijection between $G = (g_1, \dots, g_n)$ and $\{1, \dots, n\}$ by mapping g_i to i .

Denote Cayley's embedding as $\varphi : G \rightarrow S_n$ and consider the matrix $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \in G$.

Write $\varphi(A)$ as a product of disjoint cycles in S_n .

Good luck!